Module 7: Geospatial Data

Urban Informatics – Fall ‘26

Agenda

1. Announcements
2. Lecture review
3. Notebook work
4. Break
5. In-class Problem Set #7
6. Group work

Announcements

- Syllabus updated
  - Nixed smart cities lecture, but prep notes are still due!

Lecture review

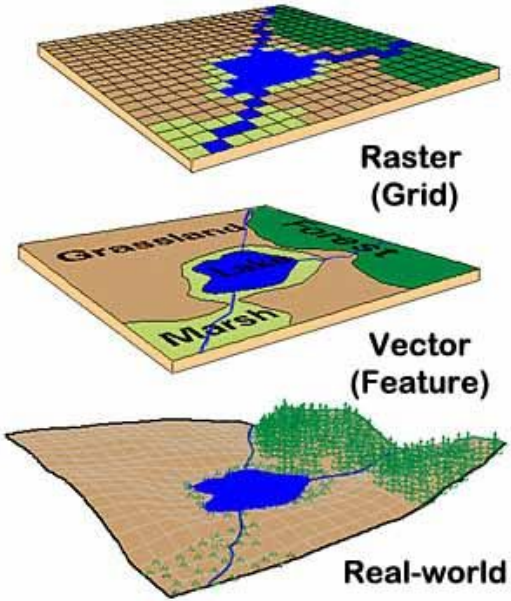
Geospatial Informatics

- GIS: Geospatial Information Systems
- Combines theory, tools, techniques from:
  - **Geography:** study of place
  - **Cartography:** study of maps and mapmaking
  - **Geodesy:** study of Earth measurement
  - **Geometry**: study of objects in mathematical space
- Geospatial data:
  - Geographic identifiers (e.g. coordinates, names)
  - Non-spatial attributes
  - Metadata

Spatial Data Types

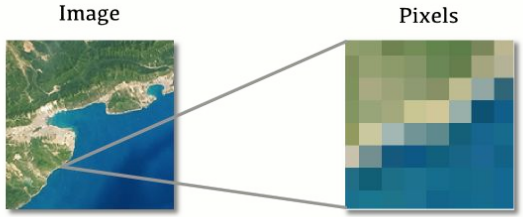
Two broad categories for representing spatially referenced data

- Raster
- Vector



Raster Data

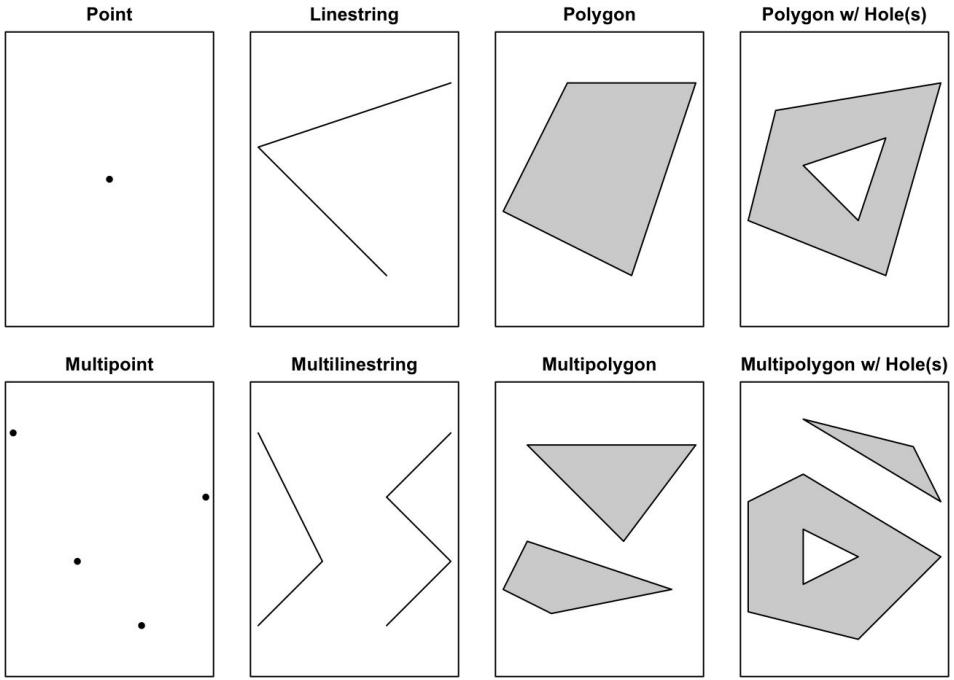
- Organized as cells in a grid
- Raster data is best suited for data which is continuously sampled at equal intervals in space.
  - e.g. photos, satellite imagery, lidar, elevation data



- A photo is just a grid of pixels, where each pixel has a centerpoint, a size, and three values for R, G, and B.
- Raster data can associate any number of values in a given grid cell.
- Fixed resolution

Vector Data

- Vector data works best for representing discontinuous features
  - lat/lon coordinates, road network, bicycle lane, administrative boundaries
- Three basic vector types or features, all based on sequences of coordinates:
  - Point
  - Line
  - Polygon
- More complex features are derived from the above:


Point
Linestring
Polygon
Polygon w/ Hole(s)
Multipoint
Multilinestring
Multipolygon
Multipolygon w/ Hole(s)

Vector Data

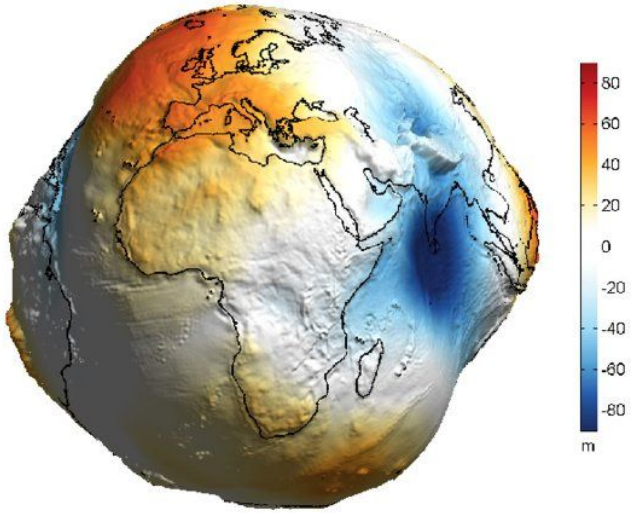
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- Three basic vector types or features, all based on sequences of coordinates:
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  - Line
  - Polygon
- More complex features are derived from the above:
- Nice for visualization because no fixed resolution,
  - Features infinitely scalable (no pixels).
  - .svg vs. .jpeg

A brief primer on Geodesy

- From NOAA:
  - Geodesy is the science of accurately measuring and understanding three fundamental properties of the Earth: its geometric shape, its orientation in space, and its gravity field
- Why do we do it?
  - The Earth is not perfectly spherical
  - All 2D representations of the surface of a globe distort the geometry of its features
  - 1° Lat != 1° Lon
- Two basic geodetic concepts allow us to navigate the surface of the earth
  - **datum** - allows us to know where we are (relative to some well-defined reference)
  - **projection** - allows us to draw our position on a map (relative to other positions)

A brief primer on Geodesy – Datums

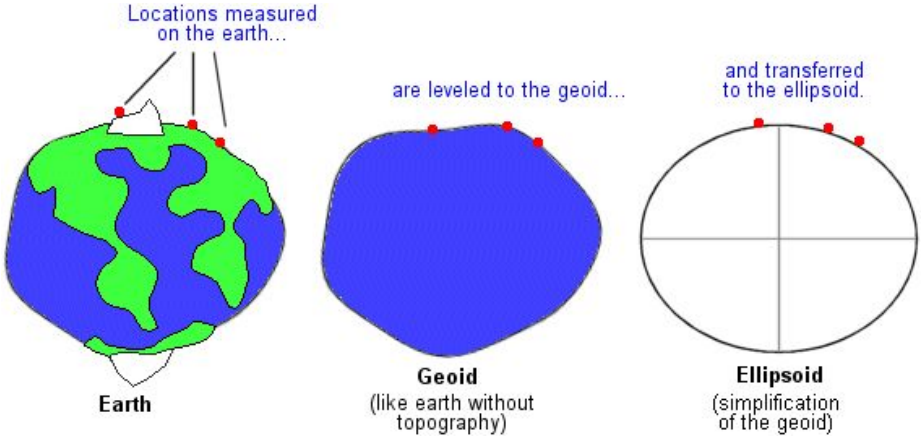
- **Geoid**: average sea-level due to gravity, rotation, etc.



Geoid height (EGM2008, nmax=500)

A brief primer on Geodesy – Datums

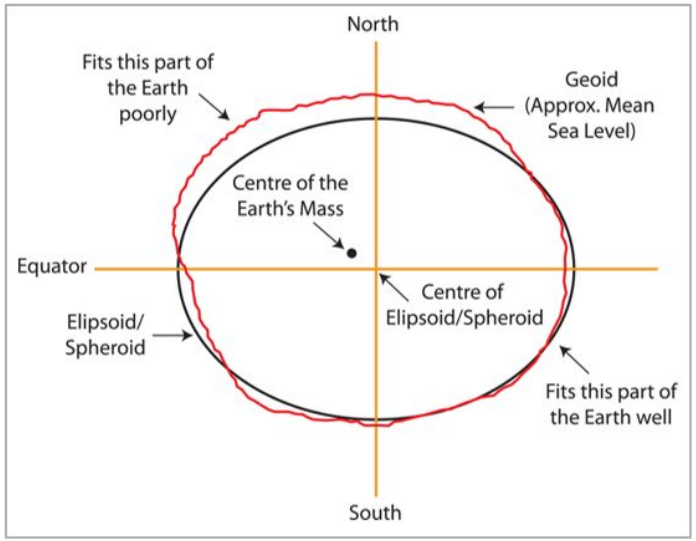
- **Ellipsoid**: the oblate spheroid of best fit

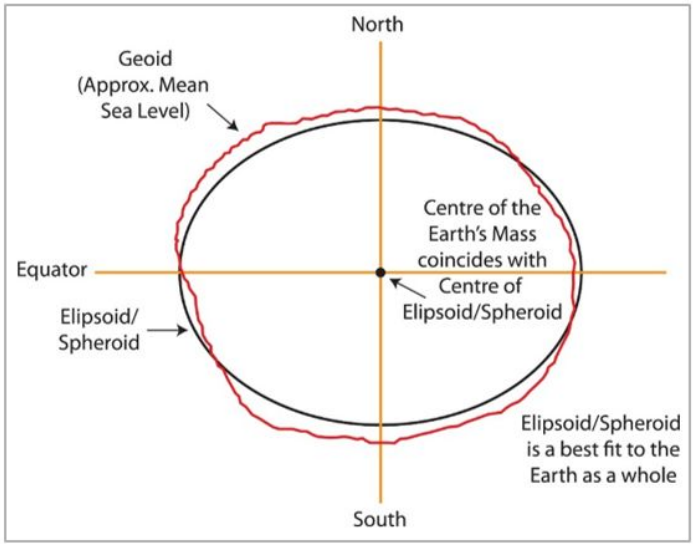


- But multiple ellipsoids are possible depending on whether you want global or local accuracy:

A brief primer on Geodesy – Datums

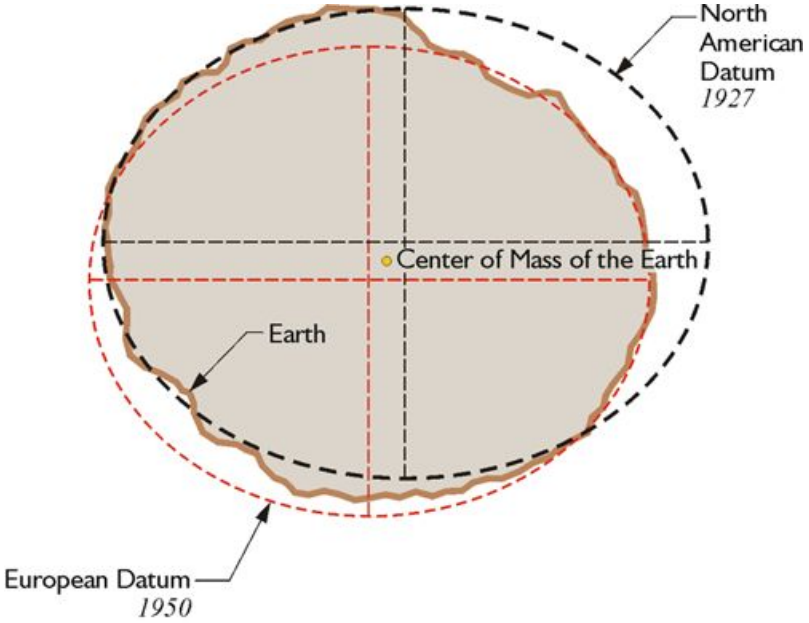
- But multiple ellipsoids are possible depending on whether you want global or local accuracy:





A brief primer on Geodesy – Datums

- A **datum** defines an ellipsoid and its orientation relative to the geoid:



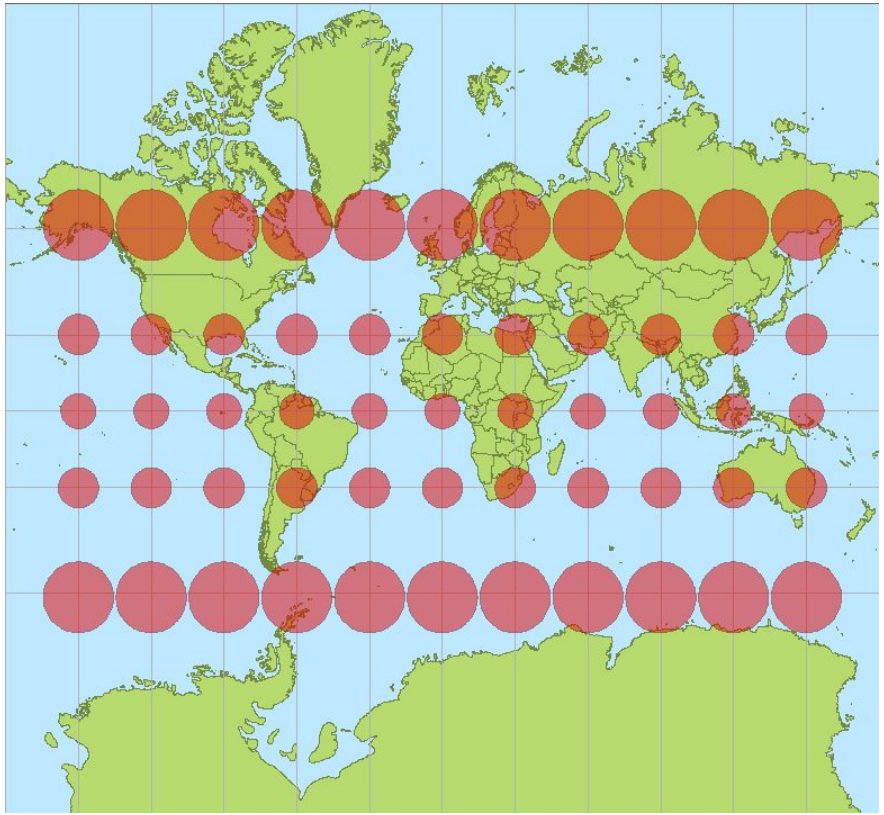
A brief primer on Geodesy – Projections

- Geoid → Ellipsoid → Datum
  - where we are on a 3D representation of Earth
- We can use projections to translate 3D coordinates to positions in two dimensional space (e.g. a map).
- Hundreds of different projections to pick from, and it's important to choose wisely.
- Why? Let's let the Organization of Cartographers for Social Equality explain:

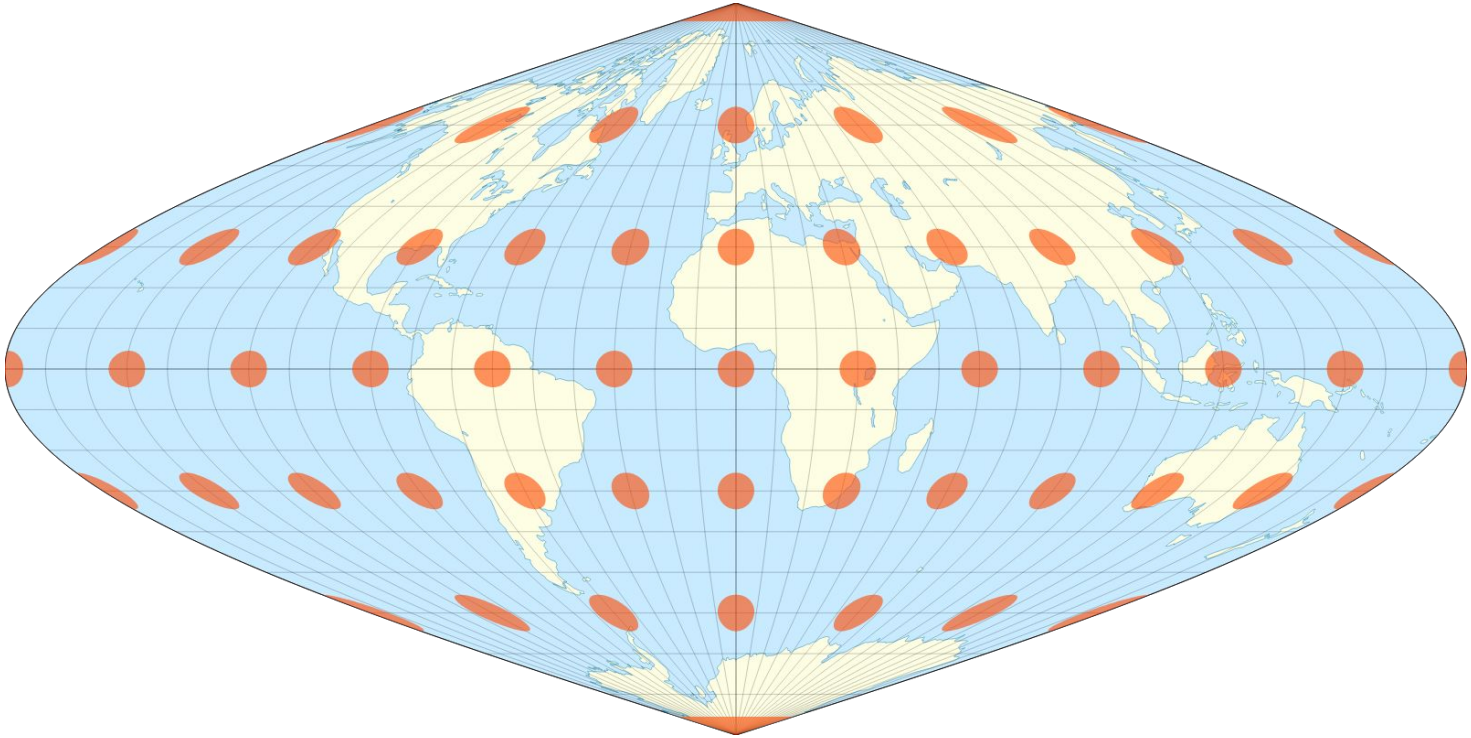


A brief primer on Geodesy – Projections

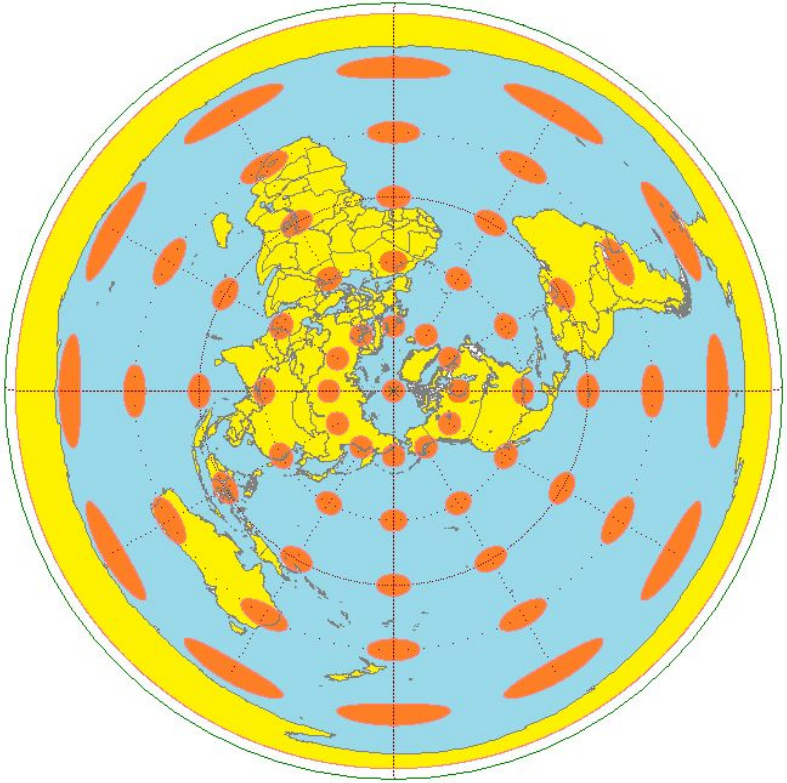
- All projections introduce visual distortion, and so all projections compromise certain kinds of accuracy in exchange for others:
  - Equal area projections preserve...area (e.g. Albers)
  - Conformal projections preserve shape (e.g. Mercator)
  - Equidistant projections preserve distance
  - Misc. others are optimized for local navigation, data collection, visual appearance, etc.



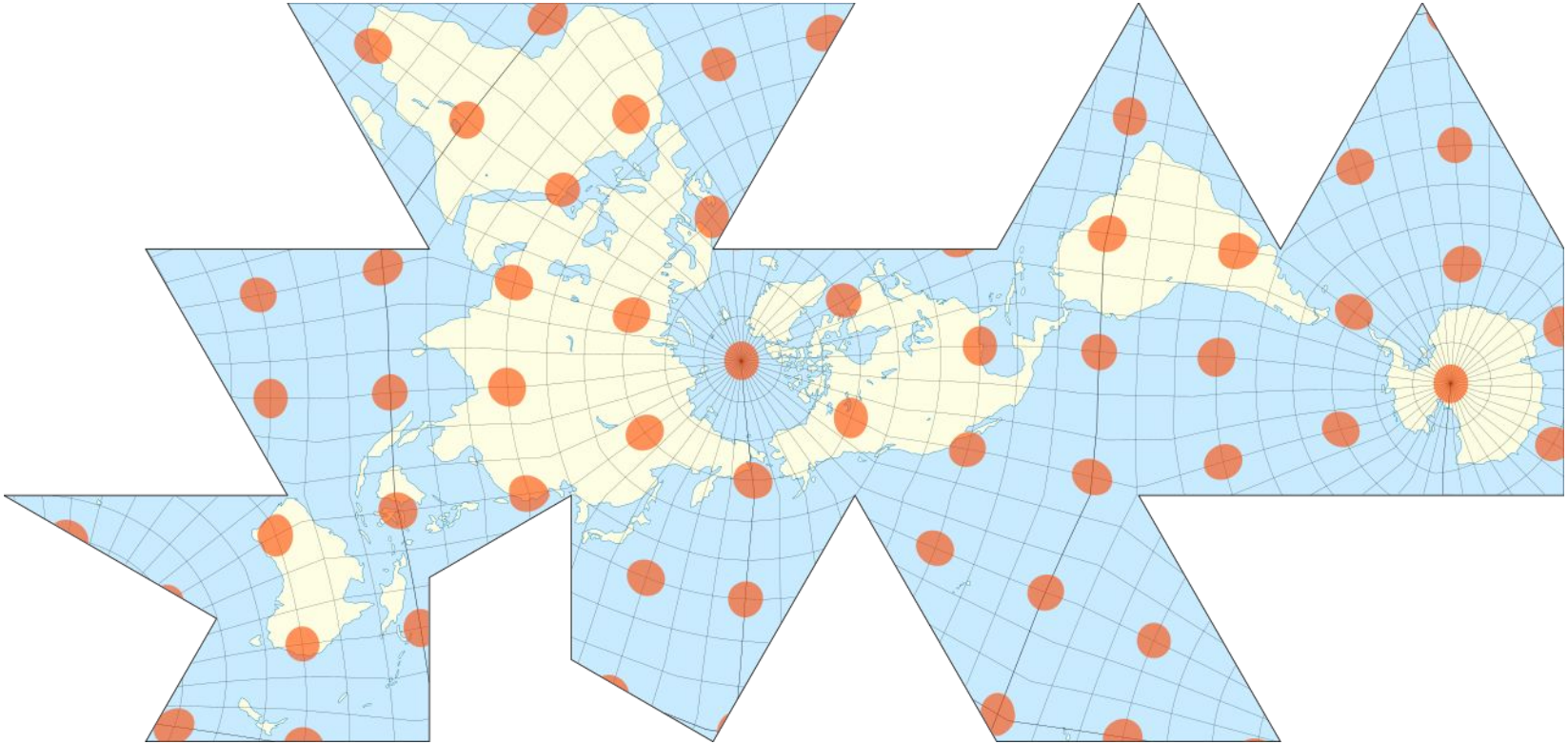
Mercator (conformal)



Sinusoidal (equal area)



Azimuthal (equidistant)



Dymaxion (compromise)

A brief primer on Geodesy – Coordinate Reference Systems

- Geographic coordinates are only useful if we know the coordinate reference system (CRS) they use. If a CRS is not explicitly defined, you'll have to guess!
- A CRS is comprised of:
  - a datum
  - a unit of measure
  - a prime meridian
  - a projection (sometimes)
- Different efforts to standardize CRS catalogue, including SRID and EPSG
- Most common CRS is WGS84, a.k.a. EPSG:4236

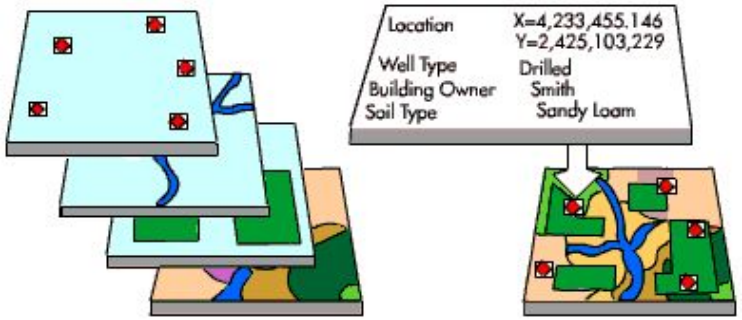
A brief primer on Geodesy – Coordinate Reference Systems

Geographic vs. Projected Coordinate Systems

- GCS uses radial coordinates (lat/lon)
  - No projection required, coordinates measured directly on the ellipsoid
  - Must use Haversine formula to compute great circle distance from degrees (assumes spherical earth) or Vincenty formulae if you need to be precise
- PCS is defined using cartesian coordinates
  - Requires projection to translate measurements to points on the ellipsoid.
  - Allows for use of standard planar measurements (e.g. feet, meters)

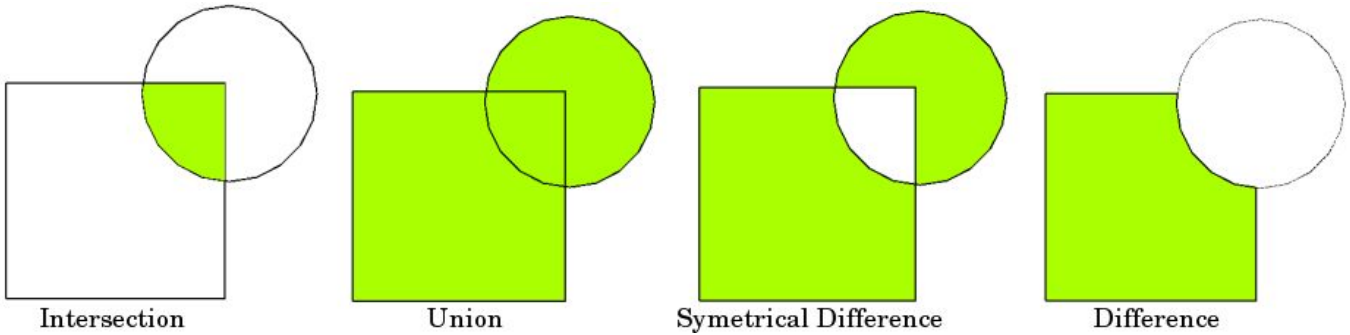
Working with Geospatial Data – Vector data edition

- Spatial joins
  - Like a merge where the "on" key is geographic
  - Requires “predicate”: e.g. within, intersects, disjoint
  - E.g. retain point geometries but get attributes of intersecting polygons



Working with Geospatial Data – Vector data edition

- Distance calculations
  - between points, along a line, nearest-neighbor, one-to-many vs. many-to-one
- Geometric manipulation
  - Centroid, buffer, dissolve, convex hull, reprojection
- Set operations (overlay)



Common geospatial file types

- Raster data typically stored as **TIFF** or **GeoTIFF**
- Web-based vector data often stored as **GeoJSON**
    - CRS is always EPSG:4326 (source)
- **Shapefile** is a proprietary format developed by ESRI. Actually a collection of files and file types:
- .shp, .dbx, .prj, .shx
- **.kml** or **.kmz** is format developed by Google and used mainly with Google Earth
- Of course, flat files (e.g. .csv) are often used for point data -- lat/lon as columns

Work on notebook prompts individually

Work on notebooks with a partner

Notebook prompt review

Break (15 minute)



Problem Set #7

Laptops and phones away please

Group work